

Chapter 1

Non-Parametric Regression

Linear Regression done last semester.

1.1 Parametric Regression (Not necessarily linear)

$$y = \alpha x_1^\beta \gamma^{x_2} \log(x + \theta x_2) + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2)$$

The form is specified and there are finite number of unknown parameters $(\alpha, \beta, \gamma, \theta, \sigma^2)$.

Data: $(x_{1i}, x_{2i}, y_i), i = 1, \dots, n$.

$$\min_{\alpha, \beta, \gamma, \theta} \sum \left[y_i - \alpha x_{1i}^\beta \gamma^{x_{2i}} \log(\dots) \right]^2 \longrightarrow \text{least squares functional}$$

1.2 Non-Parametric Regression

Non-parametric $\rightarrow$ Do not know the functional form of the dependence between x and y .

Some smoothing is done and we assume that the function

$$f(y) = f(x) + \varepsilon$$

is “smooth” (continuity & some degrees of differentiability).

We have $(x_i, y_i), i = 1, \dots, n$.

$$y = f(x) \quad (\varepsilon = 0)$$

New (x, y) should be close to the curve.

1.2.1 Objectives

- How well the estimate $(\hat{f})$ fits the observed data.
- Complexity of the model, smoothness (number of derivatives that exist); say 4th derivative is zero .

We are estimating f (parameter, infinite-dimensional) by estimate $\hat{f}$ to achieve the above objectives.

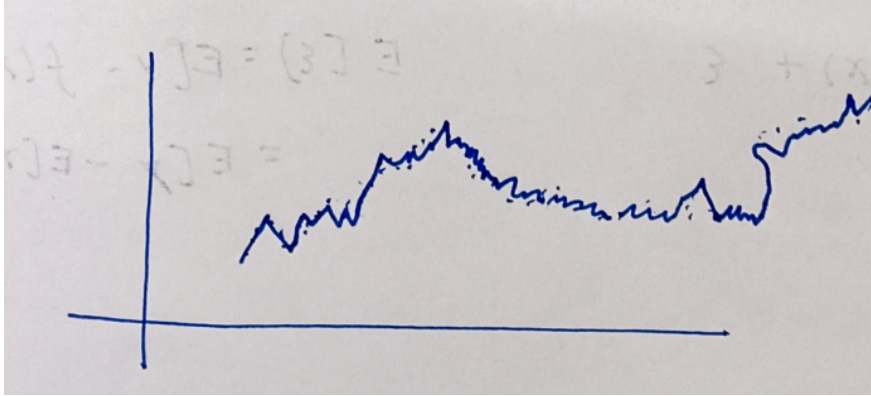


Figure 1.1

1.2.2 Least Squares Estimation by $f(x)$

Minimise $\mathbb{E}(y - f(x))^2$.

$$\begin{aligned}\mathbb{E}[y - f(x)]^2 &= \mathbb{E}[y - \mathbb{E}[y|x] + \mathbb{E}[y|x] - f(x)]^2 \\ &= \mathbb{E}[y - \mathbb{E}[y|x]]^2 + \mathbb{E}[\mathbb{E}[y|x] - f(x)]^2 \\ &\quad + 2\mathbb{E}[(y - \mathbb{E}[y|x])(\mathbb{E}[y|x] - f(x))]\end{aligned}$$

The cross term vanishes because $\mathbb{E}[y|x] - f(x)$ is non-random, and

$$\mathbb{E}[y - \mathbb{E}[y|x]] = \mathbb{E}[y] - \mathbb{E}[\mathbb{E}[y|x]] = 0.$$

The LHS is minimised as a function of f ; the 2nd term is zero when

$$\boxed{f(x) = \mathbb{E}[y | x]}$$

Check: $y = f(x) + \varepsilon$, so

$$\mathbb{E}[\varepsilon] = \mathbb{E}[y - f(x)] = \mathbb{E}[y - \mathbb{E}[y|x]] = 0. \checkmark$$

1.3 Estimating $f(x) = \mathbb{E}[Y | X = x]$

We want the conditional expectation of Y given $X = x$. We do *not* know the distribution of Y given $X = x$.

For a given x , look at the K many x_i that are “close” to x ; take the conditional distribution of Y given $X = x$ as most information is in those corresponding y_i , K-nearest neighbour regression.

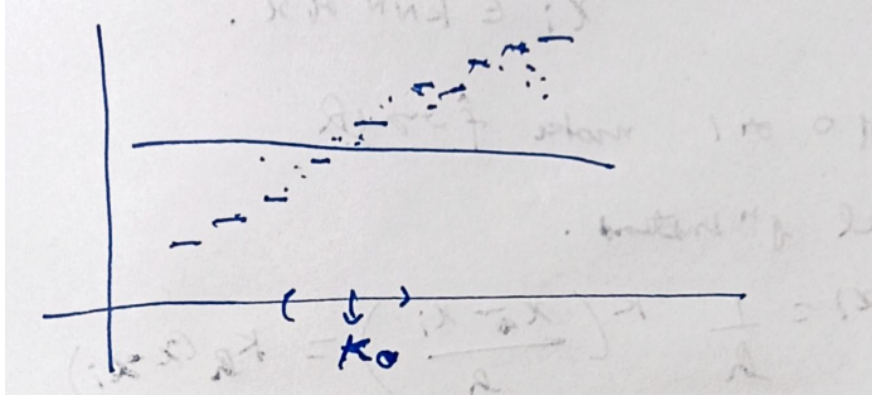


Figure 1.2: K nearest neighbour regression

1.3.1 K -Nearest Neighbour Regression

$$\{i : |x - x_i| < h\} \longrightarrow \text{average of those } y_i.$$

Local averaging: piecewise straight lines that smooth out for larger K .

$$w_i(x) = \begin{cases} 1 & \text{if } |x - x_i| < h, \\ 0 & \text{otherwise.} \end{cases}$$

$$\hat{y} = \hat{f}(x) = \frac{\sum_i w_i(x) y_i}{\sum_i w_i(x)}$$

KNN (K nearest neighborhood definition (alternative form)):

$$w_i(x) = \begin{cases} 1 & \text{if } x_i \in \text{KNN of } x, \\ 0 & \text{otherwise.} \end{cases}$$

Given $X = x$, the conditional distribution of Y assigns mass $\frac{1}{K}$ to the nearest neighbours of x_i ; $y = y_i$ with probability $\frac{1}{K}$ if $x_i \in \text{KNN of } x$.

1.3.2 Kernel Regression

Weights of 0 or 1 make $\hat{f}$ rough; use a *kernel* function instead:

$$w_i(x) = \frac{1}{h} K\left(\frac{x - x_i}{h}\right) = K_h(x - x_i)$$

$y = y_i$ with probability $\frac{w_i(x)}{\sum_j w_j(x)}$ = continuous distribution.

$$\mathbb{E}[Y|x] = \hat{f}(x) = \frac{\sum_i w_i(x) y_i}{\sum_i w_i(x)}$$